

GentlePoles: Designing Wooden Pole Actuators for Guiding People

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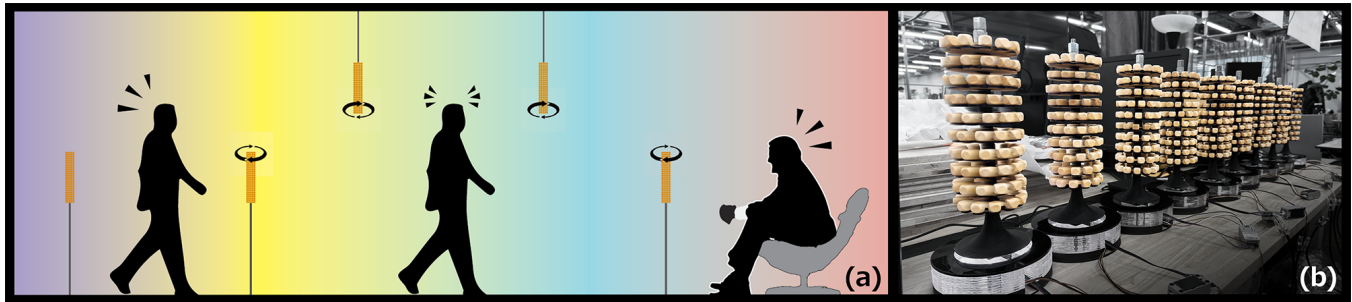


Figure 1: (a) Concept Image: Gently providing awareness and information to people through rotation. (b) Prototype Image: A wooden prototype with motorized rotational control, allowing gentle tactile interaction by hand.

Abstract

GentlePoles are wooden pole-like actuators that gently rotate to guide people without relying on text signs or staff. By arranging poles with individually controllable rotation direction and speed, the system gently and subtly directs people. For example, changes in rotation can convey messages such as "go" or "stop". In environments like airports, these poles can serve as triggers for passengers to access the information they need, while providing visual reassurance to surrounding passengers. The goal is to gently present information in public spaces with low cognitive load, addressing challenges such as staff shortages and information overload.

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1 Introduction

In public architecture, such as airports, museums, and commercial facilities, where large numbers of people pass through, signage systems using text, pictograms, color, symbols, and video are essential for directing and regulating people. However, constantly adding such information can lead to cognitive overload, making spaces difficult for travelers to navigate. Therefore, methods of presenting information that minimize cognitive load are needed. Mark Weiser argued for the importance of Calm Technology [Weiser 1995], and Hiroshi Ishii emphasized ambient media that present information as subtle cues, as demonstrated in projects such as ambientROOM [Ishii et al. 1998] and PinWheels [Ishii et al. 2001]. This study proposes GentlePoles as a method that subtly approaches the periphery of human awareness, exploring the feasibility of unconscious and gentle crowd control.

2 Gentle Poles

2.1 Concept

Gentle Poles are devices that add rotational motion to existing architectural elements such as stanchion poles and pendant lamps, subtly presenting information cues to people (Figure 1(a)). Additionally, the poles can subconsciously guide people to less crowded cafes, prompt passengers to prepare as their boarding time approaches, and create anticipation for meals that are about to be served.

In this way, this system is a device that aims to solve chronic problems in public spaces such as airports, not only by providing behavioral guidance without relying on human labor or textual signage, but also by relaxing passengers and reducing staff workload.

2.2 Prototype

First, an actuator capable of controlling the rotation direction and speed was created by combining a stepping motor (Nema 17) and motor driver (ATOM)(Figure 1(b)). The surface material used was wood, chosen for its demonstrated relaxing effects on visual, tactile, and olfactory senses[Song et al. 2021].

3 Experiment

3.1 Setup

Next, we investigated how the different directions of rotation of the wooden poles influenced participants' unconscious behavioral decisions and interpretations. Six rotating wooden poles were used to create two lanes: one with inward rotation and the other with outward rotation (Figure 2). Participants were not given any prior explanation about the test's purpose or the poles' functions; instead, they were allowed to walk freely through each lane. Finally, interviews were conducted to explore the reasons for their lane choice and the impressions they felt during the passage.

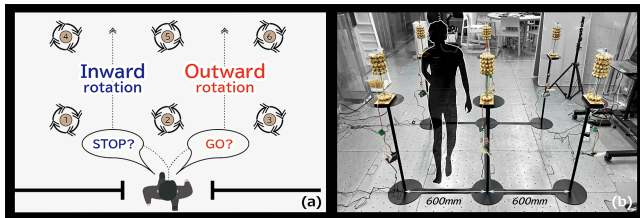


Figure 2: Experiment with two lanes and poles rotating in different directions.(a)Concept,(b)Actual experiment scene.

3.2 Results

Five undergraduate and graduate students participated in this study (three males and two females). Participants A and B reported feeling resistance against the opposite direction of their movement on the inward-rotating lane, while the outward-rotating lane facilitated forward movement. After noticing the difference in rotation direction, Participant B stated he could only choose the outwardly rotating lane that matched his movement. Participants C and D couldn't clearly explain their lane choices, and Participant E didn't notice the rotation difference until the researcher explained it.

3.3 Discussion

The interview results suggest that the direction of pole rotation influences participants' behavior. Inward rotation, opposite to movement, may subconsciously signal them to stop or hesitate. Conversely, outward rotation, aligning with movement, may encourage forward motion by providing a positive sensation (Figure 2(a)). This suggests potential for new behavioral guidance methods without relying on traditional cues like text or symbols.

However, the variability in participants' interpretations requires further investigation. Additional research is needed to understand how factors like passage width and pole spacing affect their impact. In this experiment, a passage width of 600 mm was used; when increased to 900 mm or 1200 mm, participants found it harder to perceive both poles simultaneously, potentially reducing the guidance effect. Adjustments to aisle width, pole size, and viewing angle are important considerations for future spatial design.

4 Future Work

Lead author Shimizu, an architect, aims to apply this approach not only to airports, but also to hospitals, schools, and urban spaces where gentle guidance is needed. However, more research is needed to identify scenarios where this method outperforms existing signage systems. While we used rotational motion in this experiment, we plan to explore other dynamic affordances, such as transformable mobile furniture robots [Nakanishi 2024] and studies that alter spatial perception through moving threads [Bhattacharya et al. 2024]. By integrating AI, IoT, and sensor technologies, this approach could evolve into a tool for real-time crowd management, personalized guidance, and emergency response. Gentle guidance that reduces cognitive load will be crucial in societies facing labor shortages and information overload. We will also consider cultural differences in interpretation as we further explore how gentle presentation of information influences subconscious behavior.

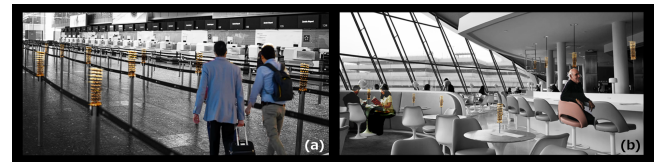


Figure 3: (a) Deployment concept for airport queue management. (b) For enhancing the waiting experience in a cafe.

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